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EXPLAINED

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India's Push Beyond E20 Fuel: Reasons and Pitfalls

• ECONOMY

India's push beyond E20 fuel: Reasons, and pitfalls



ANIL SASI

THE CENTRE last week exempted higher ethanol-petrol blends (22%-30% ethanol) from central excise duty, putting their tax treatment on a par with the 20% ethanol blend (E20) that's the standard at pumps.

The government has also proposed amendments to recognise 85% ethanol-blended fuel and 100% ethanol fuel under the Central Motor Vehicles Rules.

Both measures signal the government's intent to prepare the country's vehicular and fuel ecosystems for the next stage of ethanol adoption. But they have differing implications.

The first measure — the transition from E20 fuel to higher blends — is aimed at most petrol cars.

The second move — the draft amendment to recognise E85 and higher fuel — targets an entirely new category called flex fuel vehicles. Such vehicles are designed to operate on varying blends of petrol and ethanol but are still some way away from being widely introduced to the Indian market.

For now, it is the first proposal that has caused concern among carmakers and motorists — especially considering their experience with the initial transition to the E20 ethanol blend from E10.

The problems for consumers...

Last year, India achieved 20% ethanol blending in petrol, five years before target. E20 fuel, denoting 80 parts petrol to 20 parts of ethanol, is now the standard petrol variant nationwide.

For people with vehicles E10-certified or older, the transition to E20 came in just three years, with little warning of the pitfalls.

There are multiple problems with using a higher ethanol blend in petrol for engines not designed for them.

ENGINE DAMAGE: Ethanol has a higher content of water, which is corrosive. So, E20 or higher ethanol blends could damage the parts of the internal combustion engine (ICE). This is especially true for older engines and the ones used on two-wheelers, where high-grade aluminum or steel casts are not used for the engine block.

MILEAGE: Many customers had flagged a drop in mileage after the E10 to E20 shift. The drop could be between 5-12%, depending on when the car was manufactured. The

• HOW A FLEX FUEL CAR IS POWERED

THE ECM: Controls the fuel mixture, ignition timing, and emissions system; detects and troubleshoots problems.

FUEL FILLER: A nozzle from a fuel dispenser attaches to the receptacle on the vehicle to fill the tank.

FUEL INJECTION SYSTEM: Introduces fuel into the engine's combustion chambers for ignition.

FUEL LINE: A metal tube or flexible hose (or a combination of these) transfers fuel from the tank to the engine's fuel injection system.

FUEL PUMP: A pump that transfers fuel from the tank to the engine's fuel injection system via the fuel line.

FUEL TANK (ETHANOL/ GASOLINE BLEND): Stores fuel on board the vehicle to power the engine.

Electronic control module (ECM)

Internal Combustion Engine (spark ignited)

Fuel Injection System

Battery

Fuel Pump

Fuel Tank (ethanol/ gasoline blend)

Fuel Filler

Exhaust System

Fuel Line

Transmission

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Petroleum Ministry, however, has said that there would be no drop in mileage would be.

THE ONLY OPTION: Vehicle owners in India currently do not have an option to choose different fuels at the petrol pump, unlike their counterparts in Brazil — a country that has successfully and structurally integrated ethanol with its automobile ecosystem. Brazilian law, unlike India, also provides for a discount for higher ethanol blends.

IGNITION ISSUE: Regular ICE cars running on higher ethanol blends are hard to start on winter mornings because ethanol burns at a higher temperature than petrol.

EXPONENTIAL WORSE PERFORMANCE: While a fuel with 10% ethanol (E10) makes little difference to a car's performance, anything above is said to cause problems in engines not specifically built for it. And this performance worsens exponentially as the blending levels increase. The Petroleum Ministry has said that these concerns around higher ethanol blends are not backed by "scientific evidence".

A senior technical expert at a major automaker acknowledged that there are no conclusive studies on the long-term impact of blended fuel on non-compliant vehicles. But the expert also said it could definitely have an impact on the engine's life. Its various rubber parts, valves and piston heads, among others.

In case of this new transition too, the

Brazil's way

Brazil's ethanol programme, which kicked off in the 1970s, was in response to the uncertainties of the oil market.

At nearly every petrol pump in Brazil now, people have the option to either choose between blended petrol, which typically includes 27-32% ethanol, and E100, which is pure hydrous ethanol. Brazil also got carmakers to usher in flex fuel cars.

consumer will likely end up bearing the brunt of any drop in fuel efficiency. For older vehicles, there is the added question over vehicle damage due to the high ethanol mix. That too is left to the consumer, a representative of an auto manufacturing association said.

...and for automakers

The E25 transition will require additional engineering and validation work around engine calibration, fuel-system durability, corrosion resistance and material compatibility. All this, just months after the E20 transition.

Vehicle costs will increase and likely fall on the consumer.

Carmakers have called for greater policy clarity and alignment before higher ethanol blends can be rolled out at scale.

Beyond that, in private, they're willing to talk about some anomalies in this E20 to E25 transition.

"Some of the concerns have been conveyed, especially through (auto) industry representatives. Indications are that the transition towards a base fuel having a higher blend of ethanol (above E20) is likely to happen, alongside the progressive roll-out of the E85-E100 fuels," a senior auto industry executive told *The Indian Express*.

Oil marketing companies (OMCs) such as Indian Oil and Bharat Petroleum, too, will face challenges. A government source

involved in the process indicated that OMCs have made it clear that they can only offer two ethanol blends at any given point.

India is looking to transition from E20 fuel to higher blends to reduce its dependence on oil imports. But the plan has raised concerns

involved in the process indicated that OMCs have made it clear that they can only offer two ethanol blends at any given point.

Behind the measure

The government is looking to progressively increase ethanol blending in petrol from the current E20 level to E25, before transitioning towards flex-fuel vehicles and E85-E100 fuels.

This is an integral part of India's strategy to reduce its dependence on imported fossil fuels — especially after the experience with the West Asia conflict that drove up prices.

India currently imports nearly 88.5% of its crude oil requirement, making the economy and energy security vulnerable to geopolitical disruptions.

There's a political angle too: the agricultural lobby, especially from Maharashtra and Uttar Pradesh, is pushing hard for ethanol blends to be hiked since farmers, such as those growing sugarcane, are sitting on significant overcapacity.

The government maintains that the rollout of higher ethanol-petrol blends will only be done after proper testing and consultations. The Bureau of Indian Standards (BIS), on May 19, also notified fuel standards for these higher ethanol blends.

According to a source in the government, the move towards blends higher than E20 is "not being pushed through in a hurry and there will be enough time given for vehicles and oil companies to gear up for this transition". "Too much should not be read into (Wednesday's) notification" on the excise duty exemption, said this source.

The transition process

In the medium term, consumers may have two categories of fuel available at petrol pumps. One would be the standard fuel — E20 today, and potentially E25 if the government follows its plans. The other would be high-ethanol fuels such as E85 or E100.

The high blend would likely be priced lower. A consumer with a flex fuel vehicle can also consider the pricing before deciding which one to use.

For new vehicles, manufacturers would have to recalibrate engines and undergo fresh emissions certification and homologation, since current models have been tested and approved using E20 fuel. Homologation, the process of certifying a vehicle's roadworthiness, would need to be repeated for E25, just as it was done when E20 was introduced.

Ultimately, however, they will likely make a shift to flex fuel vehicles — a project delayed on multiple counts.

India's Push Beyond E20 Fuel: Reasons and Pitfalls

Context

Centre exempted higher ethanol-petrol blends (22–30% ethanol) from central excise duty tax treatment at par with E20.

Draft amendment proposed to recognise E85 and E100 as fuel categories under Central Motor Vehicles Rules. Both measures aim to prepare India for flex-fuel vehicles cars that run on varying blends of petrol and ethanol.

India currently imports nearly 88.5% of its crude oil requirement and reducing dependence is a key policy goal.

Ethanol Blending Programme (EBP)

E10: 10% ethanol, 90% petrol. E20: 20% ethanol, 80% petrol, currently the standard at most petrol pumps.

India achieved 20% ethanol blending in petrol (E20) last year five years before the 2025 target.

Next steps: E25 → flex-fuel vehicles (E85 and E100).

Ethanol is sourced primarily from sugarcane (Maharashtra, UP) and grain

India's Push Beyond E20 Fuel: Reasons and Pitfalls

The Ethanol Blended Petrol (EBP) Programme aimed at mixing ethanol with standard motor petrol. Operating under the ambit of the **National Biofuel Policy (NBP) 2018**, the program is driven by the Ministry of Petroleum and Natural Gas.

The program achieved a monumental milestone when India successfully hit its **20% ethanol blending target (E20) in 2025, a full five years ahead of its original 2030 deadline.**

Benefits

1. Import Cost Reduces
2. Farmers Income increases (India's farmers from Annadataas (food providers) into Urjadaatas (energy providers)).
3. Lower Emissions

India's Push Beyond E20 Fuel: Reasons and Pitfalls

National Biofuel Policy

First Generation (1G)	Edible biomass (Sugarcane, Corn, Wheat)
Second Generation (2G)	Non-edible biomass (Agri-residue, Paddy Stubble, Bamboo)
Third Generation (3G)	Algae-based
Fourth Generation (4G)	Genetically modified crops/organisms

India's Push Beyond E20 Fuel: Reasons and Pitfalls

Problems and Pitfalls

Engine Damage: Ethanol has higher water content corrosive to internal combustion engine (ICE) parts, especially in older vehicles.

Mileage Drop: E20 to E25 transition may cause 5–12% drop in mileage for existing vehicles.

Ignition Issues: Higher ethanol blends are harder to start in cold/winter conditions.

Performance: Performance worsens exponentially as ethanol blending increases beyond E20. Consumers will bear cost of vehicle modification, and **vehicle costs will increase.**

Food and Fuel Conundrum

Flex Fuel Vehicles (FFVs)

Vehicles designed to operate on any blend of petrol and ethanol.

Brazil's model: Structurally integrated ethanol with automobile ecosystem provides discounts for higher ethanol blends.

India is far from this model : limited FFV options, single fuel at pumps, homologation needs to be redone.

Bureau of Indian Standards (BIS) notified fuel standards for higher ethanol blends on May 19, 2026.

Practice Questions

Q. With reference to India's Ethanol Blending Programme (EBP), consider the following statements:

- 1. E20 fuel refers to a blend of 20% ethanol and 80% petrol.**
- 2. Flex-fuel vehicles can operate on any blend of petrol and ethanol including E85 and E100.**
- 3. Brazil's ethanol programme is considered unsuccessful as it failed to integrate ethanol with its automobile ecosystem.**

Which of the above statements is/are correct?

- (a) 1 and 2 only**
- (b) 2 and 3 only**
- (c) 1 only**
- (d) 1, 2 and 3**

Answer : (a) 1 and 2 only

Statement 3 is incorrect — Brazil's ethanol programme is considered a global SUCCESS story. Brazil has successfully and structurally integrated ethanol with its automobile ecosystem, and Brazilian law provides discounts for higher ethanol blends at petrol pumps. It is held up as the model for India to follow.

Where to apply for UPSC Exam

- **Prelims** : E10, E20, E25, E85, E100 ethanol blends; Flex Fuel Vehicles (FFV); Ethanol Blending Programme (EBP); Bureau of Indian Standards (BIS)
- **Mains** : Examine India's Ethanol Blending Programme — opportunities and challenges. / How can India reduce its dependence on crude oil imports through biofuels?
- **Interview** : Should India mandate flex-fuel vehicles given the concerns of engine damage and mileage loss? / What is India's energy security strategy?

Army Unveils New Uniform Policy, Phases Out Colonial-Era Practices

Army unveils new uniform policy, phases out colonial-era practices

Saurabh Trivedi
NEW DELHI

In a significant move aimed at modernising military traditions and reinforcing an indigenous identity, the Indian Army has introduced revised dress regulations that progressively remove residual colonial-era practices, terminology, and non-essential accoutrements.

The new policy, elaborated in the Army Uniforms-2026 Pamphlet, reflects the Army's effort to align its dress regulations with contemporary Indian values while maintaining the service's rich traditions and professional standards.

A notable feature of the revised regulations is the



The changes are aimed at enhancing uniformity, practicality and ease of implementation across the force. SPECIAL ARRANGEMENTS

introduction of a common Uniform Numbering Scheme for the Army, Navy, and the Air Force. The initiative is intended to improve clarity, interoperability, and synergy among the three services while simplifying references to various dress categories.

The Army has retained four broad categories of uniforms – Ceremonial Dress, Working Dress, Mess Dress, and Combat Dress. Unique dress numbers have been assigned to each uniform for ease of identification and administration.

Reflecting India's evolving national identity and contemporary ethos, the latest edition of the Army Uniforms Pamphlet introduces several key changes aimed at shedding remnants of colonial-era practices. Among the notable reforms are the inclusion of the traditional Bandi Jacket as part of formal civil attire, removal of the pouch belt from Mess Dress Nos. 5 and 6, making sword carriage by the Reviewing Officer optional, and discontinuing outdated terminology such as "Royal".

The Army has announced the phased withdrawal of legacy patterns such as Dress No. 3A, which will be discontinued by June 30, 2029.

Army Unveils New Uniform Policy, Phases Out Colonial-Era Practices

The New Policy

Indian Army introduced revised dress regulations under the Army Uniforms-2026 Pamphlet. Aims to progressively remove residual colonial-era practices, terminology, and non-essential accoutrements.

Reflects India's evolving national identity and contemporary ethos aligning military traditions with Indian values.

A notable feature is the introduction of a common Uniform Numbering Scheme for the Army, Navy, and Air Force.

Key Changes

Inclusion of traditional **Bandi Jacket** as part of formal civilian attire replacing colonial-era Western jacket.

Removal of the pouch belt from Mess Dress Nos. 5 and 6.

Sword carriage by the Reviewing Officer made optional.

Discontinuation of outdated terminology such as 'Royal'.

Legacy pattern Dress No. 3A will be discontinued by June 30, 2029.

Army Unveils New Uniform Policy, Phases Out Colonial-Era Practices

Four Broad Uniform Categories (Retained)

Ceremonial Dress : for formal state and military ceremonies.

Working Dress : day-to-day professional duties.

Mess Dress : formal dinners and mess functions.

Combat Dress : field and operational use.

Unique dress numbers assigned to each uniform for ease of identification and administration.

Significance

Part of India's broader decolonisation effort in national institutions : similar to renaming of Rajpath to Kartavya Path.

Common Uniform Numbering Scheme improves inter-service clarity, interoperability, and synergy among Army, Navy, Air Force.

Signals India's shift toward indigenous identity in its armed forces — consistent with Atma Nirbhar Bharat in defence.

Practice Questions

Q. With reference to the Indian Army's new uniform policy (2026), which of the following statements is/are correct?

- 1. The Army Uniforms-2026 Pamphlet introduces a common Uniform Numbering Scheme for the Army, Navy, and Air Force.**
- 2. The traditional Bandi Jacket has been included as part of formal civil attire under the new policy.**
- 3. The new policy retains five broad categories of uniforms — Ceremonial, Working, Mess, Combat, and Parade Dress.**

Which of the above statements is/are correct?

- (a) 1 and 2 only**
- (b) 2 and 3 only**
- (c) 3 only**
- (d) 1, 2 and 3**

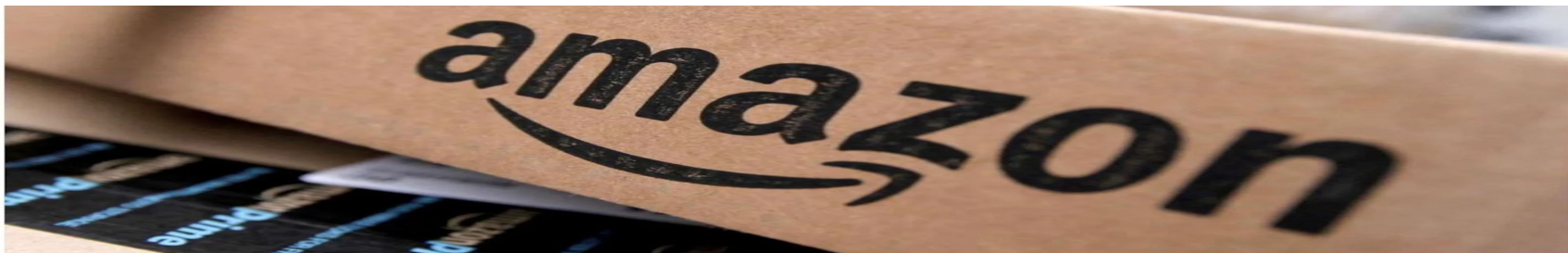
Answer : (a) 1 and 2 only

Statement 3 is incorrect — the Army retains FOUR broad categories of uniforms: Ceremonial Dress, Working Dress, Mess Dress, and Combat Dress. There is no separate 'Parade Dress' category mentioned in the new policy.

Where to apply for UPSC Exam

- **Prelims** : Army Uniforms-2026 Pamphlet; Four uniform categories; Common Uniform Numbering Scheme; Bandi Jacket
- **Mains** : Discuss India's decolonisation efforts in national institutions with examples. / What is the significance of the Indian Army's new uniform policy in the context of indigenous identity?
- **Interview** : Is decolonisation of military practices important for national identity? / What does 'Atmanirbhar Bharat' mean in the context of India's armed forces?

What is Drop Shipping, and What are the Risks for Online Shoppers?



The success of Amazon, the largest e-commerce retailer in the world, can be credited at least in part to drop shipping practices. REUTERS

What is drop shipping, and what are the risks for online shoppers?

Drop shipping allows online individuals or agents who do not hold any products to take customer orders and pass them on to other makers or sellers, inserting themselves into the transaction as middlemen; while legal in most jurisdictions, the practice can expose customers to several risks

Sahana Venugopal

While scrolling through Instagram Reels, you naturally pause when an attractive product catches your eye: a sleek laptop bag for travel, a cheap set of popular novels, an ultrafast wireless charger on sale, beautifully embroidered linen shirts, or leather-bound notebooks with custom gold foil inscriptions. Before you know it, you are entering your credit card number or UPI PIN.

Caught up in your shopping spree, you may not have checked the true origin of the product or other customers' reviews of the seller. What's more, you may not even realise that the "company" you paid is just a decorated webpage created with AI and owned by person who does not hold a single product.

This is the economy of drop shipping: the business practice that allows almost anyone with an internet connection to sell products that come from others.

Here is what you should know as a customer.

What is drop shipping?

This is a business practice where an online individual or agent — who does not hold any product — takes orders from customers and passes them on to another maker/seller who actually holds the product. The product is then delivered to the customers, either by the original maker/seller or a third-party delivery service. To put it simply, the drop shipper inserts itself in between the customer and the middleman. One or even multiple drop shippers might be in the loop, and the original makers of the product they want.

The success of Amazon, the largest e-commerce retailer in the world, can be credited at least in part to drop shipping practices. In its earliest days, Amazon did not commandeer giant international warehouses packed with products; rather,

the company was a web-based middleman that fulfilled many customers' orders by sourcing the books they wanted.

Though Amazon's product delivery model has drastically evolved, the company allows drop shipping to take place via its platform. Other channels, such as Shopify, also enable drop shipping and help users manage the logistics. Alternatively, drop shipping might take place through channels such as WhatsApp, Instagram, Facebook, and lesser-known websites.

In essence, if you have ever bought any product online, there's a chance that you placed your order through a drop shipper rather than the original seller.

How does drop shipping work?

Drop shippers can work with both domestic and international manufacturers and customers. Some try to fill an existing gap in their local market (such as fashion and home decor), some capitalise on market trends (such as the popularity of iPhone cases), and others build hype around rarer products (such as foreign luxury fashion). Most importantly, the drop shipper does not need to buy or store any of the products they are selling.

Drop shipping has become a popular source of income for both public-facing Instagram creators monetising their virality as well as private marketers looking for discreet revenue streams. What's more, a drop shipper feels like something that almost anyone with an internet connection can try out.

In addition to promoting products and setting up online storefronts, drop shippers may also buy/sell courses that reveal current market trends, customer interests, trusted wholesalers, and useful AI tools, in order to generate further profits.

Drop shipping might look logical at

first, but it is in vogue for a number of reasons.

Buyers trying to independently source foreign products from a wholesaler might be forced to order more than they need, or contend with complicated customs regulations and fees. They also may not know if the original seller is legitimate, or might be unable to communicate with them due to a language barrier.

However, placing orders through a vetted drop shipper with a user-friendly web page eases some of these worries. What's more, legitimate drop shippers regularly carry out quality assurance tests of their own, and should ideally become a customer support layer of sorts in order to resolve any complaints.

Is drop shipping legal?

Drop shipping is legal in most jurisdictions, as long as the participants involved are transparent and comply with their region's tax laws.

Instagram's community guidelines require users state that the users' Facebook Pages or Instagram professional accounts must contain product listings that are available for direct purchase from the users' website. In case buyers are being directed to another website, the user must provide the domain through which they sell, and their channels must represent the shop associated with the domain.

But now that it is possible to create entire storefronts and e-commerce websites in minutes with just some cheap domains and a few AI coding prompts, businesses can more easily present themselves as original sellers and offload customer orders to other sellers. Videos of workers sorting through

happily holding up stacks of paper that their claim are customer orders, can mislead social media shoppers into thinking they are dealing directly with a trusted seller.

Ultimately, the line separating

influencers, marketers, and sellers can blur quickly on social media. Some drop shippers also ask customers to contact them via WhatsApp, to take their communication off the app. Buyers may not know where their products are actually coming from or how their data privacy rights are being eroded. And as supply chain transparency is lost, drop shipping operations can quickly slide into a legal grey area or end up in criminal violations.

Some risks include scams targeting customers, as well as scams targeting drop shippers. Original sellers and drop shippers may both inflate product prices. Drop shippers might also work with wholesalers sending out defective/rejected products.

Furthermore, drop shippers may unknowingly promote pirated/counterfeit products. Hyper-realistic AI product images can mislead both customers and casual drop shippers.

Meanwhile, delivery timelines can be long if the product even arrives at all. The drop shipper may not be required to take responsibility for any safety or hygiene standards that have to be maintained while shipping the products.

Adding to this, multiple drop shipping factors in different countries can inadvertently lead to sanctions violations. In case of complaints, cancellations, or refund requests, customers may have no choice except to cut their losses and move on.

Other dangers include pyramid schemes to recruit more drop shippers, or even cyberstalking via phishing and hacks, as the customers' payment information is shared with multiple parties without their informed consent.

In the next time you spot a book influencer on Instagram promoting a handy new reading light, or stumble across an excited founder surrounded by sacks of customer orders, take a few minutes to do your research and find out who is really selling you the product.

What is Drop Shipping, and What are the Risks for Online Shoppers?

What is Drop Shipping?

A business practice where an online individual or agent takes customer orders and passes them on to another maker/seller who actually holds the product.

The drop shipper inserts itself as a middleman — does not hold any inventory.

The product is then delivered to the customer either by the original maker or a third-party delivery service.

Platforms like Amazon, Shopify, WhatsApp, Instagram, Facebook enable drop shipping.

How Drop Shipping Works

Drop shippers connect domestic/international manufacturers with customers.

Some fill gaps in local markets (affordable textbooks), some capitalise on trends (iPhone cases), others build hype around rare products (foreign luxury fashion).

AI tools automate customer services and harvest business insights — making drop shipping accessible to almost anyone.

Drop shippers may also buy/sell courses revealing market trends, wholesalers, and AI tools.

What is Drop Shipping, and What are the Risks for Online Shoppers?

Risks for Consumers

Product origin unknown buyers may not know where products come from or if the original seller is legitimate.

Data privacy concerns customer payment information shared with multiple parties without consent.

Counterfeit/pirated products drop shippers may unknowingly or knowingly promote fake goods.

Delivery timelines long product may not arrive at all; no accountability for safety or hygiene standards.

Scams: pyramid schemes to recruit more drop shippers, phishing, and cyber-crimes.

Sanctions violations multiple drop shipping layers in different countries can inadvertently lead to legal breaches.

Legality

Drop shipping is legal in most jurisdictions as long as participants ensure transparency and comply with tax laws.

Instagram's commerce eligibility requires professional accounts to have product listings available for direct purchase.

The line between legitimate sellers and casual dropshippers is blurred, especially on social media.

Practice Questions

Q. With reference to 'drop shipping' in e-commerce, which of the following statements is/are correct?

- 1. In drop shipping, the intermediary holds the inventory and ships it directly to the customer.**
- 2. Drop shipping is illegal in most jurisdictions due to consumer protection concerns.**
- 3. Drop shippers can expose consumers to risks such as counterfeit products, data privacy breaches, and undisclosed supply chains.**

Which of the above statements is/are correct?

- (a) 1 and 3 only**
- (b) 3 only**
- (c) 2 and 3 only**
- (d) 1, 2 and 3**

Answer : (b) 3 only

Statement 1 is incorrect — in drop shipping, the intermediary does NOT hold any inventory. The product is held by the original maker/seller and delivered directly.

Statement 2 is incorrect — drop shipping is LEGAL in most jurisdictions, as long as participants ensure transparency and comply with regional tax laws.

Statement 3 is correct — counterfeit goods, data privacy risks, and supply chain opacity are real risks for consumers.

Where to apply for UPSC Exam

Q.52) An e-commerce revenue model where the seller has control over pricing but doesn't keep products in stock and instead transfers customer orders and shipment details to a third-party supplier, who then ships the goods directly to the customer, is called:

- a) Dropshipping Model
- b) Affiliate Revenue Model
- c) Transaction Fee Revenue Model
- d) Agency Revenue Model

The 'Seven-Point IQ Opportunity' for Indian Children

The 'seven-point IQ opportunity' for Indian children

What if we could increase the IQ of India's future generations by seven points with a pivot in the understanding of early childhood development under the Anganwadi system?

For decades, India's early childhood agenda has focused on ensuring that children survive early childhood, reducing under-five mortality from 43 in 2012 to 32 in 2020 (UNICEF and World Bank data) with sufficient inputs invested toward their health, nutrition, immunisation and sanitation conditions. The Anganwadi system reflects these priorities at scale: approximately eight crore children (ages 0-6 years) receive supplementary nutrition, growth monitoring and health services, alongside counselling and home visits for their parents. However, body and brain development in the early years do not proceed on separate tracks.

Early childhood development is an ecological process in which nutrition, health, and early learning interact dynamically with each other through the body, brain, and environment. The developing brain is highly energy-intensive, consuming nearly one-fifth of the body's energy at rest. In the first year alone, grey matter volume increases by 149% and the cerebellum by 240% as millions of synapses form in regions associated with movement, language, and planning.

Middle-income evidence is clear

Paediatric research in Jamaica in the 1980s provided hard academic evidence to an idea that many cultures already understood. Researchers working with stunted children found that nutritional supplementation improved physical health alone, but adding regular psychosocial stimulation (dove-talk-play and responsive interaction) resulted in stronger cognitive gains.

Similar findings emerged in India from a birth cohort study in Vellore (250 children followed from birth to age 9) where children stunted early in life performed worst on later cognitive assessments, but those who recovered physically also showed better cognitive outcomes.



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Anganwadis can help boost cognitive development through holistic early intervention

International research on enteric dysfunction and poor sanitation has shown that nutritional inputs are mediated by the child's environment. A child's body may receive food, but infection, inflammation, toxic exposure and deprivation of stimulation can adversely influence this very nutrition. In studies in Vellore, children with iron deficiency or high lead exposure in early life, score lower on developmental and cognitive assessments, even when their height and weight appear normal. It is not simply what children eat that matters, but whether their bodies can absorb and use it. Yet, numbers on a growth chart never tell the full story. The quality of the home environment – how much they are talked to, read to, played with and emotionally supported, keeps emerging as a stronger predictor of cognitive outcomes than diet alone.

Most encouragingly, exposure to structured preschool and early stimulation has the potential to transform a generation. Children in the Vellore cohort who attend preschool regularly for 18 to 24 months, including in Anganwadi centres, are scoring seven IQ units higher on cognitive tests than those not attending preschool, even after accounting for poverty, maternal education and early stunting. In a similar Brazilian birth cohort, children who received structured preschool had higher cognition scores by eight units at age five. This is powerful evidence that stimulation and nutrition have strong, independent effects and amplify each other when both are fulfilled.

A role for Anganwadis

Indian policymakers are responding to new evidence quickly. The Anganwadi system has started to incorporate this broader understanding of holistic early childhood development into its core design. This shift is most visible in the rollout of national frameworks such as Aadharshila and Navchetana and programmes such as Poshan Bhi Padhai Bhi. They are critical inputs in redefining neighbourhood Anganwadis as vibrant early childhood education centres that nourish both mind and body. Aadharshila

strengthens play-based preschool education within centres, while Navchetana extends early stimulation into the home, equipping caregivers with simple, science-based ways to support development through everyday interactions.

In clinical practice, we offer these frameworks to parents everyday, helping to turn routine moments into opportunities for learning. We encourage the integration of Navchetana's "loving, talking and playing" into daily life while feeding, cooking or playing with children. The home lays the foundation of early language, cognition and socio-emotional skills in the first three years, and Anganwadis deepen it through structured play-based activities, storytelling and peer learning from ages three to six.

Systemic community mobilisation initiatives, such as Poshan Pakhwada conducted in April 2026 by the Ministry of Women and Child Development focus on promoting early stimulation for brain development, play-based education in the early years, and the role of parents and communities in reducing screen exposure and strengthening engagement with Anganwadis. The campaign signals that early childhood development is not confined to centres or programmes, but depends on what happens within households and neighbourhoods.

Building futures

An "ecological" approach uplifts families as much as it strengthens children. Reliable childcare gives mothers the time, the confidence and the opportunity to work, study or contribute economically. In Vellore, we have seen women trained as childcare workers gain both dignity and livelihood, creating a virtuous circle of care. When the kitchen, classroom and crèche align, children flourish, women work and communities thrive.

For a Viksit Bharat, progress cannot rest on calories alone. The physical and interactive learning and playing environments in homes, Anganwadis and communities must be as enriching as the meals they receive.

The 'Seven-Point IQ Opportunity' for Indian Children

Context and Key Claim

Early exposure to structured preschool and stimulation can increase a child's IQ by seven points — evidence from Vellore birth cohort study (CMC Vellore).

Children attending Anganwadi preschool regularly for 18–24 months scored 7 IQ units higher on cognitive tests than those not attending.

India's early childhood agenda has focused on survival and nutrition but brain development needs more: stimulation, play, and responsive caregiving.

The Anganwadi System

Approximately 8 crore children (ages 0–6 years) receive supplementary nutrition, growth monitoring, and health services through Anganwadis.

Anganwadis serve under the ICDS (Integrated Child Development Services) scheme — Ministry of Women and Child Development.

New national frameworks — Aadharshila and Navchetana — being rolled out to incorporate holistic early childhood development into Anganwadi design.

Poshan Bhi Padhai Bhi initiative — redefines Anganwadis as vibrant early childhood education centres.

The 'Seven-Point IQ Opportunity' for Indian Children

The Science of Early Brain Development

In the first year alone, grey matter volume increases by 149% and the cerebellum by 240% — millions of synapses form.

The developing brain consumes nearly one-fifth of the body's energy at rest.

Body and brain development do NOT proceed on separate tracks — nutrition, stimulation, and environment interact dynamically.

Enteric dysfunction and poor sanitation can mediate how well a child absorbs nutritional inputs — inflammation blocks nutrition uptake.

Poshan Pakhwada 2026

Conducted in April 2026 by the Ministry of Women and Child Development.

Focus: Promoting early stimulation for brain development, play-based education, role of parents and communities, and reducing screen exposure.

Signals that early childhood development is not confined to Anganwadi centres — depends on what happens within households and neighbourhoods.

Practice Questions

Q. With reference to India's Anganwadi system and early childhood development, consider the following:

- 1. Anganwadis function under the Integrated Child Development Services (ICDS) scheme of the Ministry of Women and Child Development.**
- 2. The Poshan Bhi Padhai Bhi initiative aims to redefine Anganwadis as early childhood education centres.**
- 3. According to the Vellore birth cohort study, children who attended preschool regularly scored seven IQ units higher on cognitive tests.**

How many of the above statements are correct?

- (a) Only one**
- (b) Only two**
- (c) All three**
- (d) None**

Answer : (c) All three

All three statements are factually correct as per the article.

Statement 1 — Anganwadis function under ICDS, Ministry of Women and Child Development.

Statement 2 — Poshan Bhi Padhai Bhi initiative redefines Anganwadis as early childhood education centres.

Statement 3 — Vellore birth cohort study (CMC Vellore) found 7 IQ unit advantage for preschool-attending children.

Where to apply for UPSC Exam

- **Prelims** : Anganwadi system; ICDS scheme; Poshan Bhi Padhai Bhi; Poshan Pakhwada; Aadharshila and Navchetana frameworks; Under-5 mortality trends
- **Mains** : Critically examine the role of Anganwadis in India's early childhood development framework. / How can India translate the 'seven-point IQ opportunity' into a national policy priority?
- **Interview** : What reforms would you suggest to strengthen the Anganwadi system? / Is nutrition alone sufficient for early childhood development in India?

Can India Protect Its Seafarers in the Gulf?

Can India protect its seafarers in Gulf?

How many Indian seafarers are working in the Gulf region? What happened to the three vessels attacked by the U.S. Navy? What sanctions were imposed on some ships? Does the foreign-flag status of the vessels limit India's options? Do international maritime laws protect civilian crews?

EXPLAINER

M. Kalyanaraman

The story so far

On June 12, after India summoned U.S. Embassy representatives in New Delhi and lodged its "strong protest" against U.S. missile attacks on merchant vessels carrying Indian crew members, U.S. Secretary of State Marco Rubio told External Affairs Minister S. Jaishankar that violations of the American blockade against "illicit transport of Iranian oil" would not be tolerated. India's flag captain, Mr. Rubio, Mr. Jaishankar posted on X: "I reiterated India's strong protest at U.S. attacks by the U.S. Navy in the Gulf that killed three Indian mariners. Such lethal actions against commercial shipping are not justified."

How many Indians are working on merchant vessels globally?
There are an estimated 3.5 lakh Indian seafarers manning ships across the world. The Indian government estimates that more than half of them are in active service, mostly in foreign-flagged ships.

This means one in every six seafarers working on large merchant ships worldwide is Indian. The International Maritime Organisation (IMO) estimates that some 20,000 seafarers of all nationalities are stranded on ships in the Persian Gulf region. The DG Shipping of India estimated at the start of the crisis that some 23,000 Indian seafarers are serving in the broader Gulf region in various capacities and facilities, with the UAE accounting for more than half.

What happened to the three vessels targeted by the U.S.?

On June 8, 9, and 11, the U.S. Navy fired precision munitions — Hellfire missiles on merchant vessels *Marinev*, *Settebello*, and *Jahar*. The three were tankers carrying Indian crew. While no one was hurt in the case of *Marinev* and *Jahar*, three Indian seafarers, one chief engineer, one engine fitter, and one deck cadet, died onboard *Settebello*. While *Marinev* was attacked off the coast of Oman, some 400 nautical miles from the Strait of Hormuz, the others were attacked off the coast of Shuaib in Oman, which is closer to the strait.

Videos put out by the United States Central Command (Centcom) show the engine rooms of the ships being hit as well as the steering compartments. These attacks damaged the vessels above the waterline, so water didn't rush in to sink them. But their ability to manoeuvre and manoeuvre on their own was lost.

In its press releases, the U.S. Centcom has said the crew of *Settebello* and *Marinev* refused to comply with Indian demands and all of them violated the blockade by attempting to transport Iranian oil. *Marinev* attempted to sail to an Iranian port, it claimed.

While the *Marinev* crew is reported to have informed seafarers unions that their ship was at anchor while attacked, *Settebello's* manager, EGS Marine EZE, a firm registered in Ajman, has flatly contradicted the U.S. Centcom. He has said *Settebello* was stationary for nearly 10 days in the region. It said the ship had no communication from the U.S. Navy and had no connection with Iranian oil or ports.

What sanctions were imposed on these ships?
Marinev was sanctioned by the U.S. in



The Palau-flagged tanker MT *Settebello*, carrying 24 Indian seafarers, was attacked by U.S. forces off the coast of Oman near the Strait of Hormuz. Three Indian crew members were killed in the strike. ANI



Kamlawati Devi, mother of one of three Indian seafarers killed aboard the tanker MT *Settebello*, mourns at her home in Boria, Uttar Pradesh, on June 12. REUTERS

December 2025 for Iranian connections. *Settebello* had been flagged by a U.S. non-profit advocacy organisation, United Against Nuclear Iran, for alleged Iranian links. U.S. sanctions are not binding on any other country besides the U.S., although the U.S. Treasury Department's sanctions make it difficult for the companies associated with the concerned ship to access the financial system and other business transactions. Quite often, sanctioned ships lose insurance cover, without which ports will not accept calls by the ships, and charterers will refuse to charter them.

There have been reports of non-compliance of these ships with some maritime regulations. Typically, such ships are typically given time to remedy them before the consequences of losing insurance cover can kick in.

Does the foreign-flag status of the vessels limit India's options?

They all carried foreign flags of convenience (FOCs) but had deep Indian links, including in ownership and management. In merchant shipping, ships are often registered in little-known nations with no infrastructure to rigorously supervise and monitor ships. Popular FOCs include Panama, Marshall

Islands, Liberia, and Guinea-Bissau. Such FOCs are also popular among ships transporting Russian and Iranian oil.

Yes, technically, an Indian flagged ship will provide legal sanction for possible action. The U.S., during the Iran-Iraq war, ordered many ships to change their flag to the U.S. so it could legally escort the ships in and out of the Strait of Hormuz. But the Indian Navy has acted against Houthi and Somali pirates to help Indian seafarers onboard foreign-flagged ships.

Further, *Marinev* and *Jahar* were often serving Indian ports, facilitating India's enormous oil needs. Soon after the ceasefire was announced in early April, *Marinev* headed out of the Strait of Hormuz to call on Mangalore. *Jahar* is a bitumen carrier — a special type of ship that carries tar bitumen, popular in laying Indian roads.

Were other options available for the U.S.?

In the last week of April, the U.S. Navy boarded *Tifani*, a sanctioned very large crude carrier in the Indian Ocean past Sri Lanka. The ship had an Indian crew.

This week, U.K. forces boarded *Syrtos* on the English Channel for its alleged links to drug cartels, Russian oil, and so on.

Merchant ships are defenceless. Typically, merchant ships have wireless

channels open for communication with navy by vessel and any instructions from governments and Navies and Coast Guard vessels are immediately complied with. Seafarers are trained to protect crew, cargo and the ship.

Somali pirates coming in small craft and carrying firearms could easily board and seize massive oil tankers. The only evasive action ships have against attackers is a high pressure water jet they can direct at them.

What can India do to protect Indian seafarers in the region?

India did launch Operation Sankalp to protect merchant ships against Houthi attacks. Indian Navy and Coast Guard intervened to protect four merchant ships in 2024. India did take action against Somali pirates, but the belligerents in the Persian Gulf are the U.S. and Iran. Therefore, India's options are rather limited.

Does international maritime law adequately protect civilian crews?

The nodal agency for regulating global shipping is the IMO, a United Nations agency. Though the IMO has the means to regulate the technical, commercial, environmental and safety aspects of global shipping through consensus and rule-making, it has often fallen short in protecting ships and seafarers. Quite often, national governments take unilateral action, and typical of an UN agency, the IMO has been unable to act to prevent or take remedial action.

This is where the U.S. and the United Nations Convention on the Law of the Sea (UNCLOS) come in. Its rules and mandates are extensive and are an outcome of decades of negotiations and consensus-building. UNCLOS is thorough in considering various situations including transits across straits such as the Hormuz. But many nations have not signed on to it. The U.S. refused to sign it on the grounds that UNCLOS considers the deep seabed as the commons. Iran has signed the convention but has not ratified it. There really is no overarching body for enforcement. Things are often left to the good sense of governments.

THE GIST

India has protested U.S. missile attacks on three merchant vessels carrying Indian crew in the Gulf, in which three Indian seafarers aboard *Settebello* were killed.

The U.S. says the vessels violated its blockade and were involved in transporting Iranian oil, but ship operators have disputed these claims and said some of the vessels were stationary when attacked.

With about 23,000 Indian seafarers in the Gulf region, India has limited options because the vessels were foreign-flagged, while international maritime bodies and laws have limited enforcement powers in such situations.

Can India Protect Its Seafarers in the Gulf?

Context

U.S. Navy fired precision munitions (Hellfire missiles) on merchant vessels MT Marivex, MT Settebello, and MT Jalwer — carrying Indian crew — on June 8, 10, and 11, 2026.

Three Indian seafarers killed aboard MT Settebello off the coast of Oman near the Strait of Hormuz.

India summoned U.S. Embassy representative Jason Meeks and lodged a 'strong protest'. EAM Jaishankar stated: 'Lethal actions against commercial shipping are not justified.'

There are an estimated 3.5 lakh Indian seafarers manning ships across the world — 1 in 6 seafarers on large merchant ships globally is Indian.

Flags of Convenience (FOC) — The Core Problem

MT Settebello was Palau-flagged; MT Marivex and MT Jalwer were also foreign-flagged ships with Indian crew.

FOC (Flags of Convenience): Ships registered in countries (Panama, Marshall Islands, Liberia, Guinea-Bissau) with minimal regulation, low taxes, and lax oversight.

FOC ships are popular for transporting Russian and Iranian oil — but lack diplomatic protection from the crew's home country.

Sanctioned ships lose insurance cover; charterers and banks refuse to handle cargo — financial pressure multiplier.

Can India Protect Its Seafarers in the Gulf?

What is UNCLOS?

United Nations Convention on the Law of the Sea — the primary international law governing the oceans.

Covers navigation rights, Exclusive Economic Zones (EEZ), territorial seas, and protection of civilian shipping.

India has signed but the U.S. has NOT ratified UNCLOS — limiting India's legal leverage.

IMO (International Maritime Organization) — UN agency for global shipping regulation; lacks enforcement powers.

Operation Sankalp and India's Limited Options

India launched Operation Sankalp to protect merchant ships from Houthi attacks in the Persian Gulf.

Indian Navy and Coast Guard intervened to protect four merchant ships in 2024 — also acted against Somali pirates.

Vessels were foreign-flagged — limiting India's ability to intervene unilaterally or seek legal redress.

About 23,000 Indian seafarers are serving in the broader Gulf region — UAE accounting for more than half.

Practice Questions

Q. With reference to India's seafarers and international maritime law, consider the following statements:

- 1. Flags of Convenience (FOC) allow ships to be registered in countries other than that of the ship owner, often with minimal regulation.**
- 2. India launched Operation Sankalp to protect merchant ships against Houthi attacks in the Persian Gulf.**
- 3. The United States has ratified the United Nations Convention on the Law of the Sea (UNCLOS).**

Which of the above statements is/are correct?

- (a) 1 and 2 only**
- (b) 2 and 3 only**
- (c) 1 only**
- (d) 1, 2 and 3**

Answer : (a) 1 and 2 only

Statement 3 is incorrect — the United States has NOT ratified UNCLOS. It considers UNCLOS's provisions extensively but has never ratified the convention, limiting international legal mechanisms available to India in this dispute. Statements 1 and 2 are correct.

Where to apply for UPSC Exam

- **Prelims** : UNCLOS; IMO; Flags of Convenience (FOC); Operation Sankalp; Strait of Hormuz; Exclusive Economic Zone (EEZ)
- **Mains** : Discuss India's options to protect its seafarers working on foreign-flagged vessels in conflict zones. / Examine the limitations of international maritime law in protecting civilian shipping crews.
- **Interview** : What is India's stand on the U.S. attack on merchant vessels carrying Indian crew? / How should India balance its strategic partnership with the U.S. and its duty to protect Indian citizens abroad?

Innovation is Strategic Necessity: Country Must Incubate Technologies World Uses

Innovation is strategic necessity. Country must incubate technologies world uses



N CHANDRASEKARAN

THE VEDIC hymn “*Aanobhadra krtavo yantu vishwatah*” (Let noble thoughts come to us from all directions) has guided India’s civilisational openness to knowledge. Today, it finds urgent expression in innovation.

Technology has become the currency of national sovereignty. A nation’s freedom in trade, security and diplomacy is increasingly determined by its position in critical technology value chains. For India, innovation is a strategic necessity, a bridge between aspiration and execution. It must expand access for citizens, raise competitiveness for enterprises, reduce strategic vulnerability for the nation, and create capabilities the world cannot easily replace. It is central to Viksit Bharat.

One of the most significant shifts of the last decade has been Prime Minister Narendra Modi’s sustained emphasis on making innovation a national mission. Initiatives such as Startup India, IndiaAI Mission, ANRF, IDEX, BIRAC, IN-SPaCE, Semicon India, Atal Tinkering Labs and the RDI Fund have built an institutional architecture for innovation, signalling that frontier technology is central to national development. Our higher education institutions, such as the IITs and IISc, supported by the Ministry of Education, are ready play grounds for deep-tech talent and function as powerful incubators.

India’s first innovation decade built the runway. It widened participation through startups, incubators, digital public infrastructure, research missions, student hackathons and new capital pathways. The next decade must build mastery. We are innovation-dense; we must now become innovation-deep. Progress must be measured by capabilities mastered, critical dependencies reduced, global value chains entered with strength, and lives improved at scale.

India has demonstrated exceptional strength in applying and adapting technol-

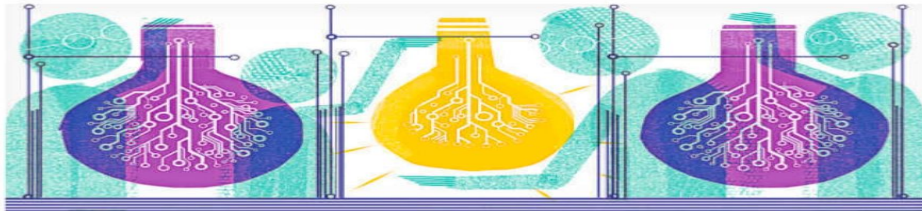


ILLUSTRATION: C R SASIKUMAR

ogy. The next frontier is origination: Owning critical IP, controlling supply-chain chokepoints, building component depth, setting standards, and creating platforms that others build upon. India’s industrial R&D intensity is low by international standards. We must move from using the world’s technology well to building technologies the world uses. In every priority sector, we should ask: Do we own the science, the design, the data, the IP, the manufacturing process, the standards and the customer access?

For India’s next phase of innovation, the relationship between policy and market forces must be sequenced and calibrated to sector maturity. In nascent frontier sectors, policy must create anchor demand and absorb certain developmental risks. IDEX proved this: Structured procurement catalysed a defence innovation ecosystem that would not otherwise have existed. As sectors mature, industry must lead, while policy fills the gaps markets leave through long-horizon research, common standards and shared infrastructure that no single player can fund.

As India focuses on technology innovation for strategic sovereignty, we need to keep in mind that it does not mean technological autarky. Given India’s starting point, its national interests are best served through deep partnerships with like-minded nations offering complementary strengths. These are not concessions of ambition. India is a design principle for reaching the frontier faster,

We must also confront technology’s double edge. AI can compress decades of progress in diagnostics and governance. Deployed without careful design, it can also disrupt work, amplify bias and deepen inequality

squarely in India’s enlightened self-interest. Strategic sovereignty means collaborating from confidence, not dependence. The India-France Year of Innovation is a template: Build mastery openly, selectively, based on mutual strength.

India’s large enterprises have a significant role to play in driving innovation beyond being first customers of the startup ecosystem. It is much more comprehensive: Investing in R&D, co-innovating with startups, integrating startups and MSMEs structurally into supply chains, making strategic investments in startups, and deploying engineering rigour, quality systems and global market access in service of the broader ecosystem.

At Tata Group, as we deepen our footprint in semiconductors, artificial intelligence, defence, clean energy and digital infrastructure, we are making the startup and MSME ecosystem integral to our growth. In every complex domain, the unit of innovation is not a product; it is an ecosystem. A semiconductor strategy needs design, materials, packaging and customers moving in concert. An electric vehicle needs a battery, charging network, software and financing to move together. Large companies that build with this understanding pull an entire ecosystem forward.

India cannot build mastery everywhere. The discipline is this: Apply rigorous criteria. What are the core socio-economic problems that need to be addressed — access to quality healthcare and educa-

tion, sustainability, energy and food security? Which domains have irreducible domestic demand, providing a captive proving ground? Where do India’s talent and cost structure create advantages that others cannot replicate? Where does import dependence create genuine strategic exposure, becoming a lever an adversary could use in a crisis? Where is a technology young enough for near-parity competition? India’s goal: To identify the domains where the world cannot easily operate without us. For this, we need to concentrate innovation energy.

We must also confront technology’s double edge. AI can compress decades of progress in diagnostics and governance. However, deployed without careful design, it can also disrupt work, amplify bias and deepen inequality. India’s model must be net positive by design, augmenting capability, creating new jobs, and ensuring that the transition does not burden the weakest sections. The transition must be planned and managed with empathy through structured reskilling and redeployment, while ensuring that India does not lag in competitiveness.

Bharat Innovates 2026, organised by the Ministry of Education in France, is a declaration that India’s most ambitious innovators are ready for the world’s most demanding stage. More importantly, it is a collaboration bridge connecting Indian startups and research institutions with global investors, corporates, universities, research institutions and markets. Its promise is not only that the world will see what India is building, but that global partners will ask: What can we build with India?

To young innovators, researchers and entrepreneurs, the conditions have never been better. India is the world’s most concentrated laboratory for consequential innovation, where the hardest problems of human development and technological self-reliance must be solved together at a billion-person scale. The innovator who solves for India’s problems solves for much of the world. That is not a burden. India is the most powerful innovation brief on earth.

India’s innovation story will be written by a generation that chose mastery over mere participation, depth over dispersal, purpose over convenience. That generation is ready. The moment is now.

The writer is chairman, Tata Sons

Innovation is Strategic Necessity: Country Must Incubate Technologies World Uses

Core Argument (N. Chandrasekaran, Chairman, Tata Sons)

Technology has become the currency of national sovereignty — a nation's freedom in trade, security, and diplomacy is determined by its position in critical technology value chains.

India must move from using the world's technology well to building technologies the world uses.

The next frontier is origination: Owning critical IP, controlling supply-chain chokepoints, building component depth, setting standards, and creating platforms that others build upon.

Strategic sovereignty means collaborating from confidence, not dependence.

India's Innovation Ecosystem

Startup India, IndiaAI Mission, ANRF, iDEX, BIRAC, IN-SPACE, Semicon India, Atal Tinkering Labs, RDI Fund — institutional architecture for frontier technology.

IITs and IISc: Ready playgrounds for deep-tech talent and powerful incubators.

iDEX (Innovations for Defence Excellence): Structured procurement catalysed a defence innovation ecosystem — policy creating anchor demand in nascent sectors.

Bharat Innovates 2026 (Paris): Declares India's most ambitious innovators are ready for the world's most demanding stage.

Practice Questions

Q. With reference to India's innovation and technology ecosystem, consider the following:

- 1. iDEX (Innovations for Defence Excellence) aims to catalyse a defence innovation ecosystem through structured procurement.**
- 2. ANRF (Anusandhan National Research Foundation) was established to fund scientific research in India.**
- 3. India's industrial R&D intensity is currently at par with global standards such as South Korea and Israel.**

Which of the above statements is/are correct?

- (a) 1 and 2 only**
- (b) 2 and 3 only**
- (c) 1 only**
- (d) 1, 2 and 3**

Answer : (a) 1 and 2 only

Statement 3 is incorrect — the article explicitly states that "India's industrial R&D intensity is low by international standards." South Korea spends ~4.9% and Israel ~5% of GDP on R&D, while India spends only ~0.65%. India is far below global benchmarks. Statements 1 and 2 are correct.

Where to apply for UPSC Exam

- **Prelims** : iDEX; ANRF; Startup India; IndiaAI Mission; IN-SPACe; Bharat Innovates 2026; Technology value chains; Strategic sovereignty
- **Mains** : Innovation is a strategic necessity for India — critically examine. / Discuss the role of public institutions in building India's deep-tech ecosystem.
- **Interview** : What does 'technology sovereignty' mean for India? / How should India balance openness to global technology with the need to build indigenous capabilities?